

Optimal Mexican Rice Experiment

Introduction

This experiment aims to determine the optimal combination of broth, garlic, and tomato sauce in Mexican rice for enhanced flavor and texture. By manipulating the quantity of broth, tomato sauce, and garlic, the experiment seeks to identify which combination yields the most palatable and visually appealing dish. For this experiment, I choose three factors: garlic, chicken broth, and tomato sauce. I chose these factors because they are three very prominent ingredients in this recipe. The primary response variable will be the "Overall Quality" of the rice, which includes assessments of texture, flavor, and visual appeal. This will be measured using a taste test conducted by a panel of judges who will score the rice based on these criteria. I would use a scale of 1 to 10 to assess how each rice tastes. Since the "Overall Quality" of the rice is rated using a 1 to 10 scale, it is considered ordinal data. In order to reduce variability, I used the same pan and measured each ingredient to ensure the cooking conditions and ingredient proportions were consistent across all trials. In addition, for each trial, I used the same heat on the stove, cooked the rice for the same amount of time.

Each of the three factors in the design has two levels:

Factor	+	-
Garlic	Garlic	No Garlic
Chicken Broth	1 cup	½ cup
Tomato Sauce	1 cup	½ cup

To test these factors, I designed a 2^3 full factorial design with 8 runs. I set up the following design:

Run	Garlic	Chicken Broth	Tomato Sauce
1	-	-	-
2	-	-	+
3	-	+	-
4	-	+	+
5	+	-	-
6	+	-	+
7	+	+	-
8	+	+	+

This design is effective because it includes only three factors, each with two levels, making a full factorial design practical. This approach provides a complete set of data that can be used to determine the significance of the factorial effects.

Data Collection

To collect the data, I cooked in my kitchen using an electric stove top, a timer, and various measurement tools. I used the same brand of rice, tomato sauce, and chicken broth for every trial.

For each run, I took the measured ingredients and applied them to the same heat of level 4 for 25 minutes. First I took $\frac{1}{2}$ tbsp of oil and poured it on the pan. Followed by putting $\frac{1}{2}$ cup of onions to saute for 5 minutes at heat 6. I then added 1 cup of rice with $\frac{1}{2}$ tbsp of each seasoning. I would then lower the heat to level 4, and place in the desired amount of chicken broth and tomato sauce

depending on the run. I set the timer for 25 minutes and let the rice cook. I then made all my roommates rate the rice out of 10 and came up with an accurate rating.

Analysis

After I collected this data, I analyzed my results in R Studio. I used a full factorial regression model including all main effects (Garlic, Tomato_Sauce, and Chicken_Broth), all two-way interactions, and the three-way interaction. The model is expressed as:

$$\begin{aligned} \text{Rating} = & \beta_0 + \beta_1(\text{Garlic}) + \beta_2(\text{Tomato_Sauce}) + \beta_3(\text{Chicken_Broth}) + \\ & \beta_{12}(\text{Garlic} \times \text{Tomato_Sauce}) + \beta_{13}(\text{Garlic} \times \text{Chicken_Broth}) + \\ & \beta_{23}(\text{Tomato_Sauce} \times \text{Chicken_Broth}) + \beta_{123}(\text{Garlic} \times \text{Tomato_Sauce} \times \text{Chicken_Broth}) + \varepsilon \end{aligned}$$

This full factorial regression model is appropriate because I am studying three categorical factors (Garlic, Tomato_Sauce, and Chicken_Broth), each at two levels. By including all main effects, two-way interactions, and the three-way interaction, the model captures not only the individual effects of each factor but also how the factors may work together to influence the rice rating.

Using a full factorial design ensures that I can investigate possible synergies or antagonistic interactions between ingredients, which is important in cooking experiments where ingredient combinations often have non-additive effects. Additionally, since the experiment was relatively simple to conduct and involved only eight runs, it was feasible to fit the full model without omitting important interaction terms.

Although the model perfectly fits the data (resulting in a saturated model with no degrees of freedom for residual error), this is acceptable given the experimental context and the small number of observations.

The output for the regression analysis looks as follows:

Coefficient	Coefficient Estimate
Intercept	7.5
GarlicYes	0.5
Tomato_SauceLow	1.5
Chicken_BrothLow	-3.5
GarlicYes: Tomato_SauceLow	-3.0
GarlicYes: Chicken_BrothLow	-0.5
Tomato_SauceLow : Chicken_BrothLow	-3.5
GarlicYes: Tomato_SauceLow : Chicken_BrothLow	3.0

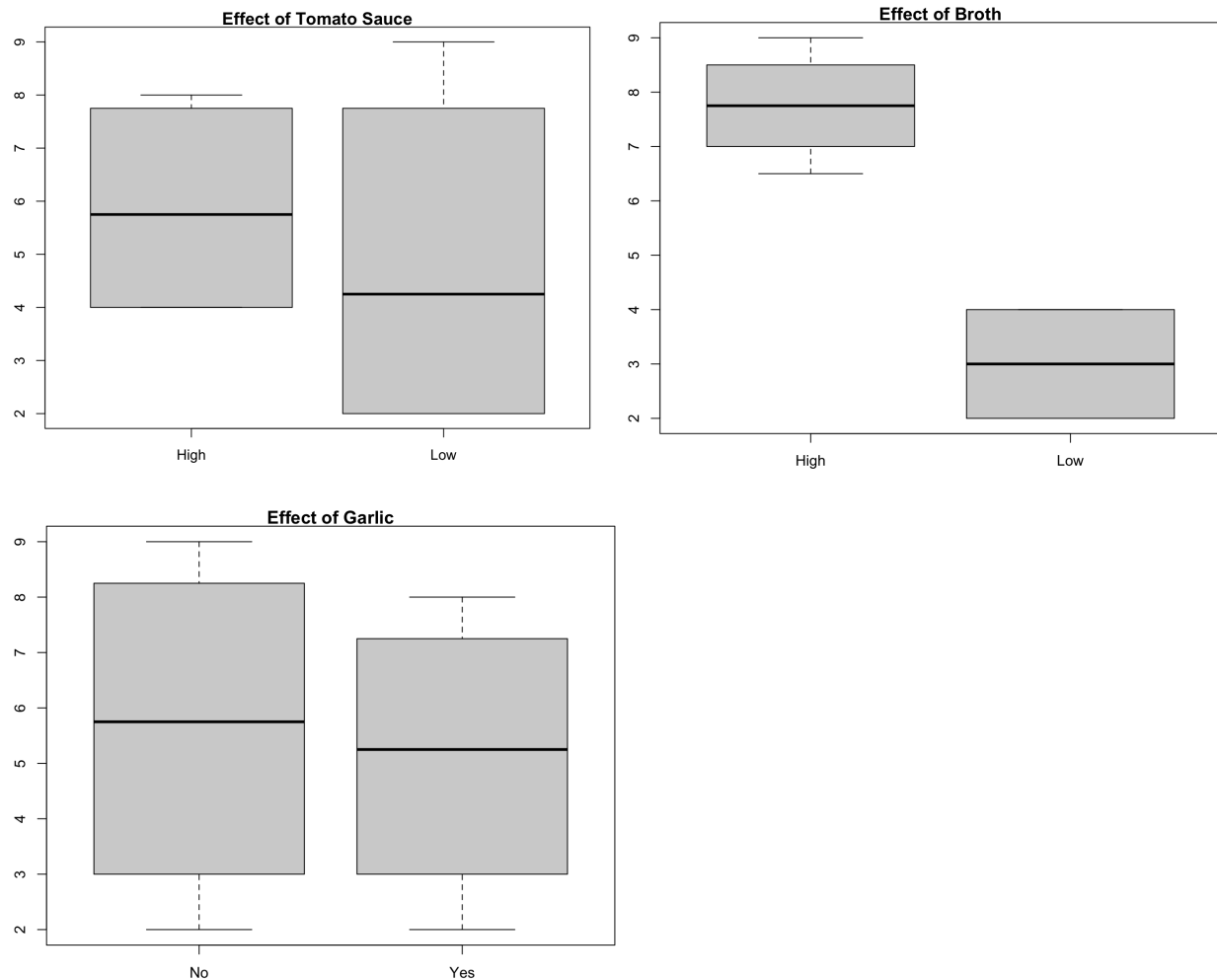
In this model, the intercept of 7.5 represents the predicted rice rating when Garlic is "No," Tomato_Sauce is "High," and Chicken_Broth is "High" — the baseline levels.

The main effect of Garlic being "Yes" increases the predicted rating by 0.5 points, while using a "Low" level of Tomato_Sauce increases the rating by 1.5 points. In contrast, using a "Low" level of Chicken_Broth decreases the rating by 3.5 points.

The interaction terms reveal that combinations of factors further influence the rating: for example, having both Garlic = "Yes" and Tomato_Sauce = "Low" reduces the rating by an additional 3.0 points. The three-way interaction between Garlic, Tomato_Sauce, and Chicken_Broth increases the rating by 3.0 points when all three non-baseline levels are present.

Since the model perfectly fits the data (saturating all degrees of freedom), no residual error remains for formal hypothesis testing. However, the estimated coefficients still provide insight into which factors and combinations had the largest impact on rice ratings.

To further explore the potential impact of each factor individually, I created boxplots showing the distribution of Ratings for the different levels of Garlic, Tomato_Sauce, and Chicken_Broth.



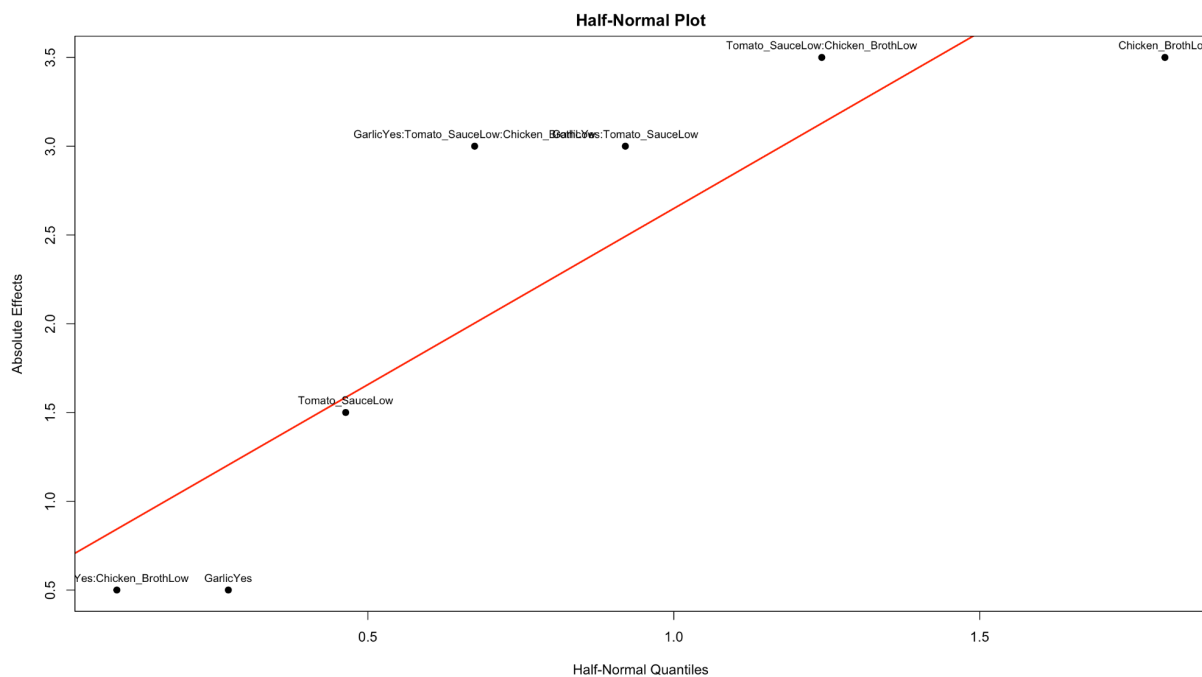
While the sample size is limited, the boxplots still offer a useful visual indication of how the factors may have influenced the Rating. Most appear to have minimal effect, but the boxplot for Chicken Broth reveals a clear and substantial difference, with higher ratings consistently associated with the high broth level. This suggests that Chicken Broth may have a practically important effect, even if it was not statistically significant in the later formal analysis using Lenth's method.

In order to formally assess which effects were large enough to be considered significant, I applied Lenth's method. I began by extracting the effect estimates from the fitted full factorial regression model (excluding the intercept) and computing their absolute values. These were used to calculate an initial estimate of variability, s_0 , defined as 1.5 times the median of the absolute effects. Next, I removed any effects greater than $2.5 \times s_0$ to form a trimmed set, from which I recalculated a more robust estimate of variability, the pseudo standard error (PSE). Using this PSE, I computed standardized effect estimates (tPSE values) by dividing each absolute effect by the PSE. Finally, I compared these tPSE values against a critical value from Lenth's method (IER = 2.3). Since no effect exceeded this threshold, none of the main effects, two-way interactions, or the three-way interaction were deemed statistically significant. Here's the output:

Coefficient	Estimate	Absolute Effect	tPSE	Significant (IER = 2.3)
GarlicYes	0.5	0.5	0.111	No
Tomato_Sauce Low	1.5	1.5	0.333	No
Chicken_BrothLow	-3.5	3.5	0.777	No
GarlicYes:Tomato_SauceLow	-3.0	3.0	0.666	No
GarlicYes:Chicken_BrothLow	-0.5	0.5	.111	No
Tomato_SauceLow:Chicken_BrothLow	-3.5	3.5	.777	No
GarlicYes:Tomato_SauceLow:Chicken_BrothLow	3.0	3.0	.666	No

Therefore, according to Lenth's method, no effects were statistically significant in this experiment. This result is not unexpected given that the model is saturated and the sample size is small, limiting the ability to detect significant effects even if they exist.

To visually assess the magnitude of the effects, I created a half-normal plot of the absolute effect estimates:



Although none of the effects were statistically significant according to Lenth's method (using $IER = 2.3$), the half-normal plot reveals that `Chicken_BrothLow`, its interaction with `Tomato_SauceLow`, and the three-way interaction with `GarlicYes` visibly deviate from the fitted line. This visual evidence suggests these effects may be practically relevant and merit further investigation, despite not reaching formal significance thresholds.

Results:

Based on the results of the regression analysis, the fitted model for this data is:

$$\begin{aligned} \text{Rating} = & 7.5 + 0.5(\text{GarlicYes}) + 1.5(\text{Tomato_SauceLow}) - 3.5(\text{Chicken_BrothLow}) - \\ & 3.0(\text{GarlicYes} \times \text{Tomato_SauceLow}) - 0.5(\text{GarlicYes} \times \text{Chicken_BrothLow}) - \\ & 3.5(\text{Tomato_SauceLow} \times \text{Chicken_BrothLow}) + \\ & 3.0(\text{GarlicYes} \times \text{Tomato_SauceLow} \times \text{Chicken_BrothLow}) \end{aligned}$$

The goal of this experiment was to figure out the optimal combination of ingredients to make the best Mexican rice. By plugging in all possible factor level combinations into the fitted regression model, the combination of Garlic = "No," Tomato_Sauce = "Low," and Chicken_Broth = "High" produced the highest predicted rating of 9.0. Because my model is saturated, the fitted values from the model exactly match the ratings I collected. The optimal factor levels, along with the two-factor and three-factor interaction levels that maximize the rice rating, are summarized below:

Factor	Optimal Level
Garlic	-
Tomato Sauce	-
Chicken Broth	+
Garlic : Tomato Sauce	-
Garlic : Chicken Broth	-
Tomato Sauce: Chicken Broth	+
Garlic : Tomato Sauce: Chicken Broth	+

Conclusion:

This experiment successfully applied a full factorial design to investigate how garlic, tomato sauce, and chicken broth levels affect the overall quality of Mexican rice. Using regression analysis, data visualization, and Lenth's method, I found that none of the individual factors or interactions were statistically significant at the selected significance level. The half-normal plot and model coefficients suggested that Chicken_BrothLow and the interaction Tomato_SauceLow : Chicken_BrothLow deviated noticeably from the general noise trend, pointing to their potential influence on overall ratings. This implies that not only does the level of chicken broth matter on its own, but its effect is altered when combined with tomato sauce. In contrast, Tomato_SauceLow alone closely followed the expected pattern of insignificant effects, suggesting its influence may depend on other ingredient combinations. The optimal factor combination based on the fitted model was Garlic = "No," Tomato_Sauce = "Low," and Chicken_Broth = "High," resulting in the highest predicted rice rating of 9.0. However, due to the small sample size and the saturated nature of the model, these results should be interpreted cautiously. Future experiments with additional replications and a broader range of factor levels would help validate these findings and provide a more reliable assessment of the best combination for producing high-quality Mexican rice.

R Programming Input and Output

Data Table Input:

```
rice_data = data.frame(  
  Rating = c(8, 4, 6.5, 2, 7.5, 4, 9, 2),  
  Garlic = c("Yes", "Yes", "Yes", "Yes",  
             "No", "No", "No", "No"),  
  Tomato_Sauce = c("High", "High", "Low", "Low",  
                   "High", "High", "Low", "Low"),  
  Chicken_Broth = c("High", "Low", "High", "Low",  
                    "High", "Low", "High", "Low")  
)  
print(rice_data)
```

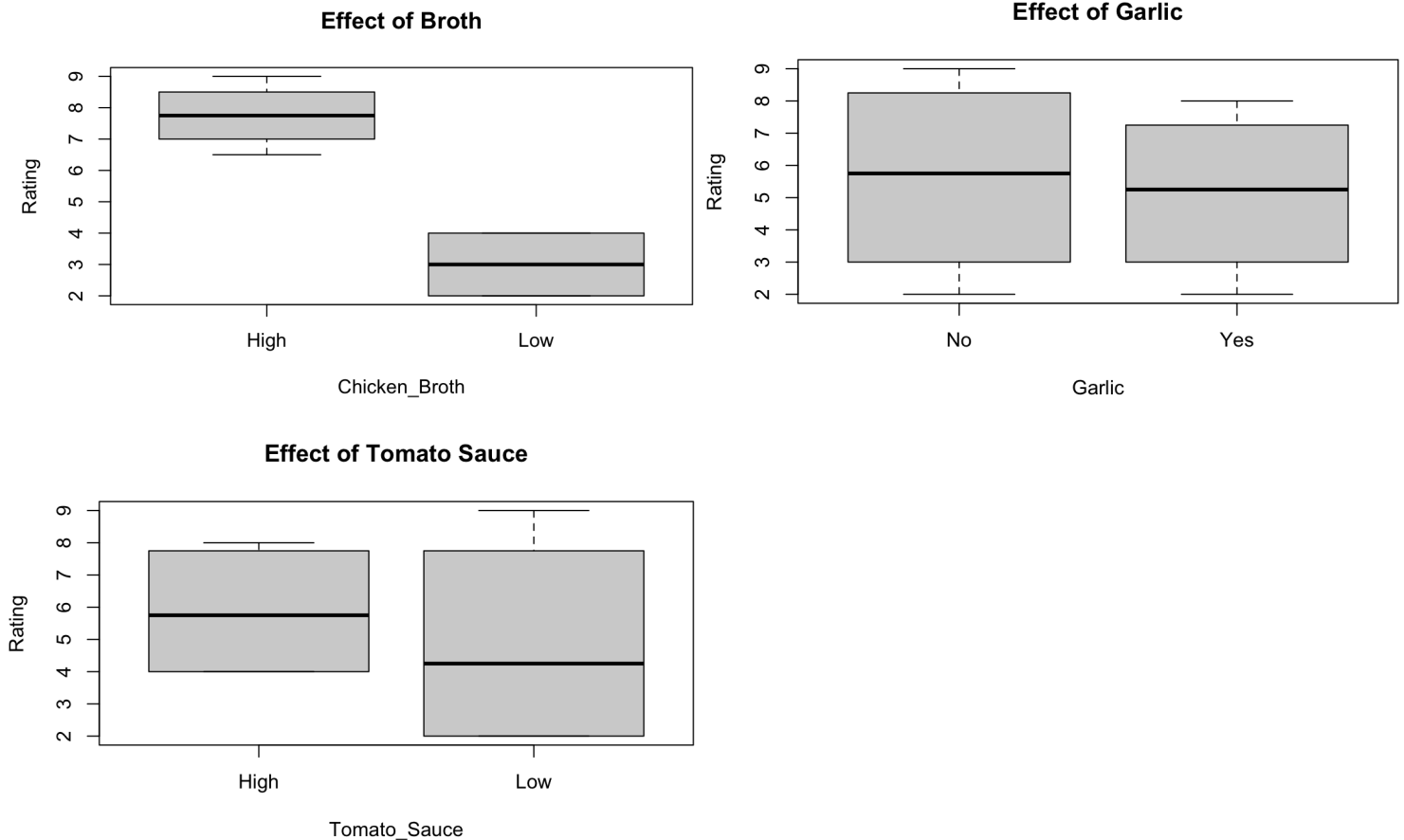
Data Table Output:

	Rating	Garlic	Tomato_Sauce	Chicken_Broth
1	8.0	Yes	High	High
2	4.0	Yes	High	Low
3	6.5	Yes	Low	High
4	2.0	Yes	Low	Low
5	7.5	No	High	High
6	4.0	No	High	Low
7	9.0	No	Low	High
8	2.0	No	Low	Low

Box Plot Input:

```
boxplot(Rating ~ Garlic, data = rice_data, main="Effect of Garlic")  
boxplot(Rating ~ Chicken_Broth, data = rice_data, main="Effect of Broth")  
boxplot(Rating ~ Tomato_Sauce, data = rice_data, main="Effect of Tomato Sauce")
```

Box Plot Output:



Regression Analysis Input:

```
Rating ~ Garlic + Tomato_Sauce + Chicken_Broth +  
  Garlic:Tomato_Sauce + Garlic:Chicken_Broth  
  + Tomato_Sauce:Chicken_Broth +  
  Garlic:Tomato_Sauce:Chicken_Broth  
  
model_full = lm(Rating ~ Garlic * Tomato_Sauce *  
  Chicken_Broth, data = rice_data)  
print(model_full)
```

Regression Analysis Output:

```
Coefficients:  
  
      (Intercept)      7.5  
      GarlicYes      0.5  
      Tomato_SauceLow      1.5  
      Chicken_BrothLow     -3.5  
      GarlicYes:Tomato_SauceLow     -3.0  
      GarlicYes:Chicken_BrothLow     -0.5  
      Tomato_SauceLow:Chicken_BrothLow     -3.5  
      GarlicYes:Tomato_SauceLow:Chicken_BrothLow      3.0
```

Lenth's Method Input:

```
effects = coef(model_full)[-1]  
abs_effects = abs(effects)  
s0 = 1.5 * median(abs_effects)  
trimmed_effects = abs_effects[abs_effects < 2.5 * s0]  
PSE = 1.5 * median(trimmed_effects)  
tPSE = abs_effects / PSE  
results = data.frame(  
  Effect = names(effects),  
  Estimate = effects,  
  Abs_Effect = abs_effects,  
  tPSE = tPSE  
)  
print(results)
```

Lenth's Method Output:

	Effect
GarlicYes	GarlicYes
Tomato_SauceLow	Tomato_SauceLow
Chicken_BrothLow	Chicken_BrothLow
GarlicYes:Tomato_SauceLow	GarlicYes:Tomato_SauceLow
GarlicYes:Chicken_BrothLow	GarlicYes:Chicken_BrothLow
Tomato_SauceLow:Chicken_BrothLow	Tomato_SauceLow:Chicken_BrothLow
GarlicYes:Tomato_SauceLow:Chicken_BrothLow	GarlicYes:Tomato_SauceLow:Chicken_BrothLow
	Estimate Abs_Effect tPSE
GarlicYes	0.5 0.5 0.1111111
Tomato_SauceLow	1.5 1.5 0.3333333
Chicken_BrothLow	-3.5 3.5 0.7777778
GarlicYes:Tomato_SauceLow	-3.0 3.0 0.6666667
GarlicYes:Chicken_BrothLow	-0.5 0.5 0.1111111
Tomato_SauceLow:Chicken_BrothLow	-3.5 3.5 0.7777778
GarlicYes:Tomato_SauceLow:Chicken_BrothLow	3.0 3.0 0.6666667

Effect Significance Input:

```
results$Significant = abs(results$tPSE) > ier.crit  
print(results$Significant)
```

Effect Significance Output:

```
[1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE
```

Half Normal Plot Input:

```
library(faraway)
```

```
thetaloc = function(I) {  
  p = (0.5 + 0.5 * ((1:I) - 0.5) / I)  
  qnorm(p)  
}
```

```
effects_abs_sorted = sort(abs_effects)  
effect_labels = names(abs_effects)[order(abs_effects)]
```

```
x = thetaloc(length(effects_abs_sorted))  
plot(x, effects_abs_sorted,  
     main = "Half-Normal Plot",  
     xlab = "Half-Normal Quantiles",  
     ylab = "Absolute Effects",  
     pch = 19,  
     abline(lm(effects_abs_sorted ~ x), col = "red", lwd = 2))
```

```
text(x, effects_abs_sorted, labels = effect_labels, pos = 3, cex = 0.8)  
print(effect_labels)
```

Half Normal Plot Output:

